Monthly Intelligence Report

Edition 001 — The Rhône Valley Corridor

Where the energy transition is outrunning the grid. Inaugural edition · April 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED

7,898

996 EHV · 6,902 HV

GRID LINES MAPPED

~12,400

~105,000 km coverage

MC SIMULATIONS

78.9 M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 78.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 469,404 output data points with full uncertainty quantification across 7,898 substations and ~12,400 grid lines spanning ~105,000 km.

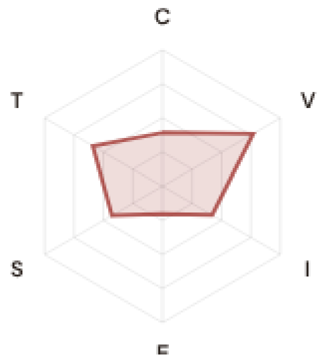
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering France's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Poste Marne 032

Marne, Grand Est · 400 kV

0.455	Medium	0.158	32th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

C Continuity: 0.392 / 0.30 (131%)
I Infrastructure: 0.421 / 0.25 (168%)
S Saturation: 0.428 / 0.20 (214%)
V Voltage: 0.769 / 0.10 (769%) ⚠
E Economic: 0.203 / 0.10 (203%)
T Transition: 0.591 / 0.05 (1181%)

MODIFIERS

R3 Consequence: 1.088 ↑ amplifies
R4 Graph Criticality: 0.969 ↓ dampens
R6a Restoration: 0.989 ↓ dampens
R6b Seismic: 1.012 ↑ amplifies
R7 Digital Readiness: 0.998 → neutral

This month's spotlight — automatically selected for April 2026 — is **Poste Marne 032** in Marne, Grand Est. With an R_final of 0.455 (Medium band, 32th fleet percentile), its risk profile is dominated by the T (Transition) component at 1181% of its weight ceiling, followed by V (Voltage) at 769%. Modifiers collectively shift R_base by 4.5%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that France Numérique / France 2030 digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

2166

High/Critical + R7 < median

AVG R7 MODIFIER

1.0143

Range [0.99, 1.05]

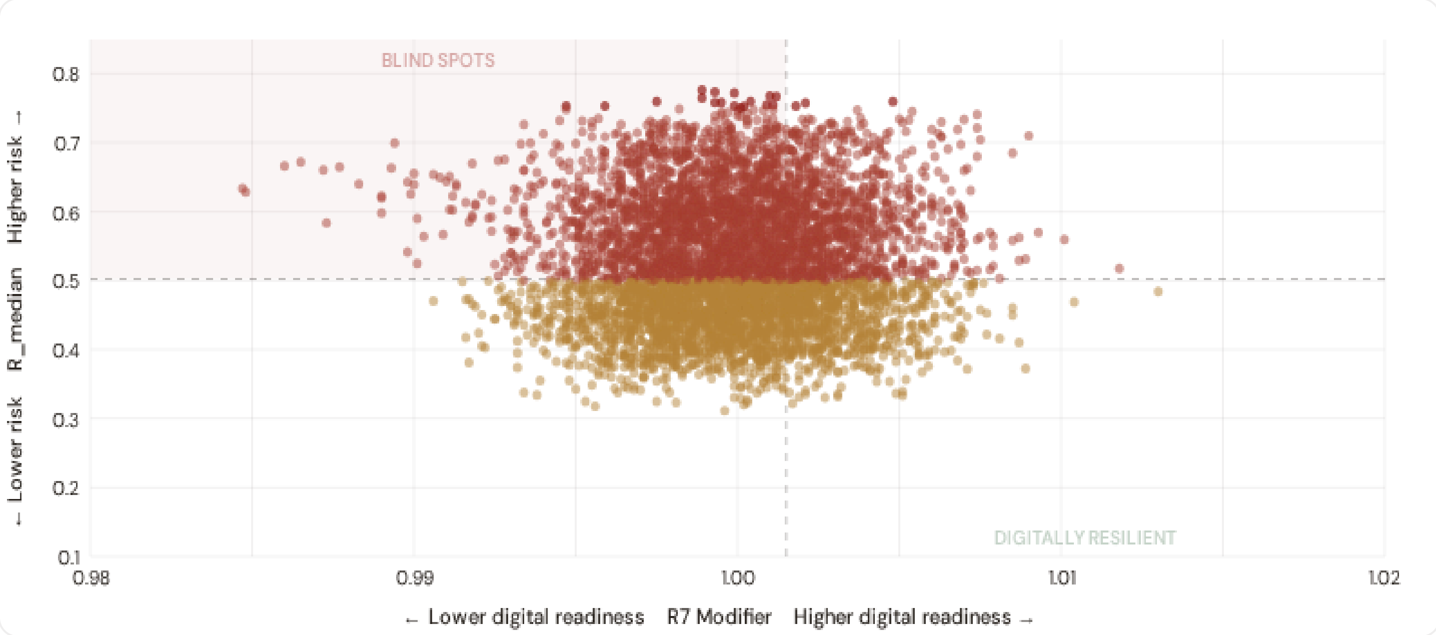
HIGH-CONSEQUENCE SUBS

2,563

32.5% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical — Dashed lines = fleet medians

The cyber-physical matrix identifies **2166 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0015). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Occitanie (621), Nouvelle-Aquitaine (572), Provence-Alpes-Côte d'Azur (402)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating strong digital infrastructure coverage but concentrated in the northern urban grid.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: €15–30/kWh

Heavy industry, petrochemical, large manufacturing — highest economic loss per interruption

1,590 substations (20.1%) High/Critical: **942** (59.2%) Avg R: **0.512** Avg E: **0.581** / 0.10

Low 102 Med 546 High 866 Crit 76

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: €8–15/kWh

Diversified industry, logistics, food processing — significant continuity requirements

1,884 substations (23.9%) High/Critical: **872** (46.3%) Avg R: **0.450** Avg E: **0.476** / 0.10

Low 293 Med 719 High 872 Crit 0

🏪 Commercial / SME-Dense

Est. VoLL: €3–8/kWh

Service economy, retail, tourism — moderate individual impact but high aggregate exposure

1,666 substations (21.1%) High/Critical: **738** (44.3%) Avg R: **0.503** Avg E: **0.290** / 0.10

Low 2 Med 926 High 738 Crit 0

🌾 Light / Agricultural / Rural

Est. VoLL: €1–3/kWh

Sparse economic activity, agricultural land — lower VoLL but higher social vulnerability

2,758 substations (34.9%) High/Critical: **1468** (53.2%) Avg R: **0.511** Avg E: **0.141** / 0.10

Low 0 Med 1290 High 1465 Crit 3

Value of Lost Load (VoLL) context: ACER's 2023 estimate for France places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **942 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Île-de-France (542), Auvergne-Rhône-Alpes (278), Hauts-de-France (216)** — regions where industrial corridors depend on uninterrupted power for continuous processes (steel, glass, automotive). A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 7.0%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Département Digital Readiness Ranking

Départements ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify départements combining digital exposure with high grid stress.

WORST DIGITAL READINESS — TOP 15 PROVINCES

Longer bar = higher R7 = better digital infrastructure.

1	Haute-Corse ⚠ high R	0.9894
2	Corse-du-Sud ⚠ high R	0.9913
3	Loir-et-Cher	0.9991
4	Indre	0.9993
5	Oise	0.9994
6	Alpes-de-Haute-Provence ⚠ high R	0.9994
7	Nord	0.9995
8	Finistère ⚠ high R	0.9995
9	Vaucluse ⚠ high R	0.9996
10	Bas-Rhin	0.9996
11	Dordogne ⚠ high R	0.9996
12	Haute-Garonne ⚠ high R	0.9996
13	Bouches-du-Rhône ⚠ high R	0.9996
14	Vosges	0.9996
15	Deux-Sèvres ⚠ high R	0.9996

Département-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Haute-Corse** has the lowest average R7 (0.9894), while **Isère** leads at 1.0507 — a spread of 0.0613 across the R7 range. **40 départements** combine below-median digital readiness with above-median grid risk, making them priority targets for France 2030 / France Relance / PPE digitalisation investment. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (April 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0143, median = 1.0015; 0 substations classified HIGH cyber-exposure; 2166 blind-spot substations (High/Critical risk + below-median R7); 17 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging départements where DESI improvements (broadband rollout, smart meter deployment, OT upgrades) translate into measurable R7 shifts. The next material R7 update is expected when Eurostat publishes the 2025 DESI sub-indices (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **35 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

WEEKLY / MONTHLY

QUARTERLY / ANNUAL

35
95 variables

5
High-frequency feeds

22
Regular refresh cycle

STATIC / DERIVED
8
Standards + scenarios

Data Source Registry — April 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
RTE-N	RTE Schéma Décennal	MONTHLY	TSO zone	2 vars
DREAL	DREAL (Directions Régionales)	CONTINUOUS	Département	1 var
ODRE	ODRÉ Registre National (OpenDataSoft)	QUARTERLY	Commune	5 vars
SDES	SDES (Service des Données Statistiques)	QUARTERLY	Département	1 var
BPI	Bpifrance Innovation	QUARTERLY	Département	1 var
CRE	CRE (Commission de Régulation de l'Énergie)	ANNUAL	Département	8 vars
INSEE	INSEE (Institut National de la Statistique)	ANNUAL	Département	8 vars
ADEME	ADEME (Agence de la Transition Écologique)	ANNUAL	Département	4 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
MTEF	Ministère de la Transition Écologique	ANNUAL	Région	2 vars
CEREMA	Cerema (Centre d'études risques)	ANNUAL	Département	1 var
DGFIP	DGFiP Fiscal Statistics	ANNUAL	Département	1 var
IRSN	IRSN Nuclear Safety	ANNUAL	Département	1 var
ONPES	ONPES / DREES Poverty Observatory	ANNUAL	Département	2 vars
ENEDIS	Enedis Open Data	ANNUAL	Département	2 vars
ANCT	ANCT Observatoire des Territoires	ANNUAL	Département	2 vars
CRE-M	CRE Monitoring Report	ANNUAL	DSO-level	2 vars
CGET	CGET/ANCT Spatial Planning	ANNUAL	Département	1 var
OFCE	OFCE Economic Research	ANNUAL	Région	1 var
ADEME-R	ADEME Régions	ANNUAL	Région	1 var
BRGM	BRGM Geological Survey	STATIC	Département	3 vars
D10	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars

DIM	Dimovski et al. (2025)	STATIC	Municipal	3 vars
IEEE-1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
COG	COG INSEE Département Registry	STATIC	Département	1 var
ISO-9223	ISO 9223 Corrosion	DERIVED	Département	1 var
RTE	RTE Open Data (éCO2mix)	HOURLY	Bidding zone	3 vars
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	2 vars
MF-OM	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars
D-OM	Open-Meteo / ERA5	HOURLY	~1 km	1 var

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Weekly (2 sources) — OSM diffs + weather reanalysis

OSM Overpass · Open-Meteo/ERA5

Monthly (3 sources) — Grid load + procurement data

ODRÉ Registry · RTE éCO2mix · RTE SDDR

Quarterly (2 sources) — DER registry + business data + macro indicators

SDES · Bpifrance

Annual (20 sources) — Core grid quality + demographics + environment

CRE · INSEE · ONPES · Enedis · MTEF · ADEME · DGFIP · EEA · Eurostat · DESI · ANCT · CRE Monitoring · CGET · OFCE · ADEME-R · IRSN · Open-Meteo · JRC · DREAL · Cerema

Static (8 sources) — Standards + scenarios + seismic hazard

Dimovski · IEEE/IEC/CIGRÉ · Copernicus CMIP6 · GFZ · BRGM · ISO 9223 · COG/INSEE · ENTSO-E

C.2 Fleet Score Snapshot

FLEET SIZE
7,898
substations scored

MEDIAN R
0.502
P5=0.250 · P95=0.697

HIGH + CRITICAL
4020
50.9% of fleet

HIGH CONFIDENCE
100%

7888 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 397 (5.0%) Medium 3481 (44.1%) High 3941 (49.9%) Critical 79 (1.0%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **7,898 substation scores remain unchanged** this edition. The fleet median R of 0.502 (mean 0.495) reflects a distribution skewed toward the lower bands, with 5.0% in Low and 44.1% in Medium. The first material score changes are expected when CRE publishes the annual TIQE indicators (affecting C1–C4 and R6a) and when ODRÉ refreshes the Registre National des Installations (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across P0, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	§2, §4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	§5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio)	§5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	§5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	§5
F6	ENHANCED	R7 Digital Readiness — Département-level DESI model	§5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeo	§2, §8
L2	ENHANCED	E2 Innovation Enrichment — HRST + startup density	§6
L3	ENHANCED	V_socio Fiscal Enrichment — DGFIP + ANCT + energy price	§5
L4	ENHANCED	R3 Demographic Shift Amplifier — INSEE net migration	§5
L5	DATA	95/95 variables operational (100%). 35 data sources total.	§8, §12
G1	DATA	d02 ODRÉ upgraded to quarterly registry — Département-level DER registry	§12
G2	DATA	d05 BRGM upgraded to live API — Département-level geological data	§12
G3	DATA	d06 OSM upgraded to Overpass API — 7,898 real substations	§12
G4	NEW	R6b Network Topology modifier — centrality + ring analysis from OSM graph	§5
G5	NEW	Network & Topology layer (H): 7 variables — RTE Schéma Décennal, ring analysis	§12
G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs	§12

SECTION D — MONTHLY DEEP-DIVE

The Rhône Valley Corridor: Where Nuclear Reliance Meets Energy Transition Pressure

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Rhône Valley corridor · **002** DER saturation hotspots (Occitanie/PACA) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan–Rural economic impact disparity · **005** Nuclear fleet dependency and transition risk · **006** Infrastructure age vs Linky rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Département Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the Rhône Valley, Why Now

RHÔNE VALLEY SUBSTATIONS

1152

315 EHV · 837 HV

HIGH BAND

825 + 59

76.7% of corridor

CORRIDOR MEDIAN R

0.561

Fleet median: 0.502

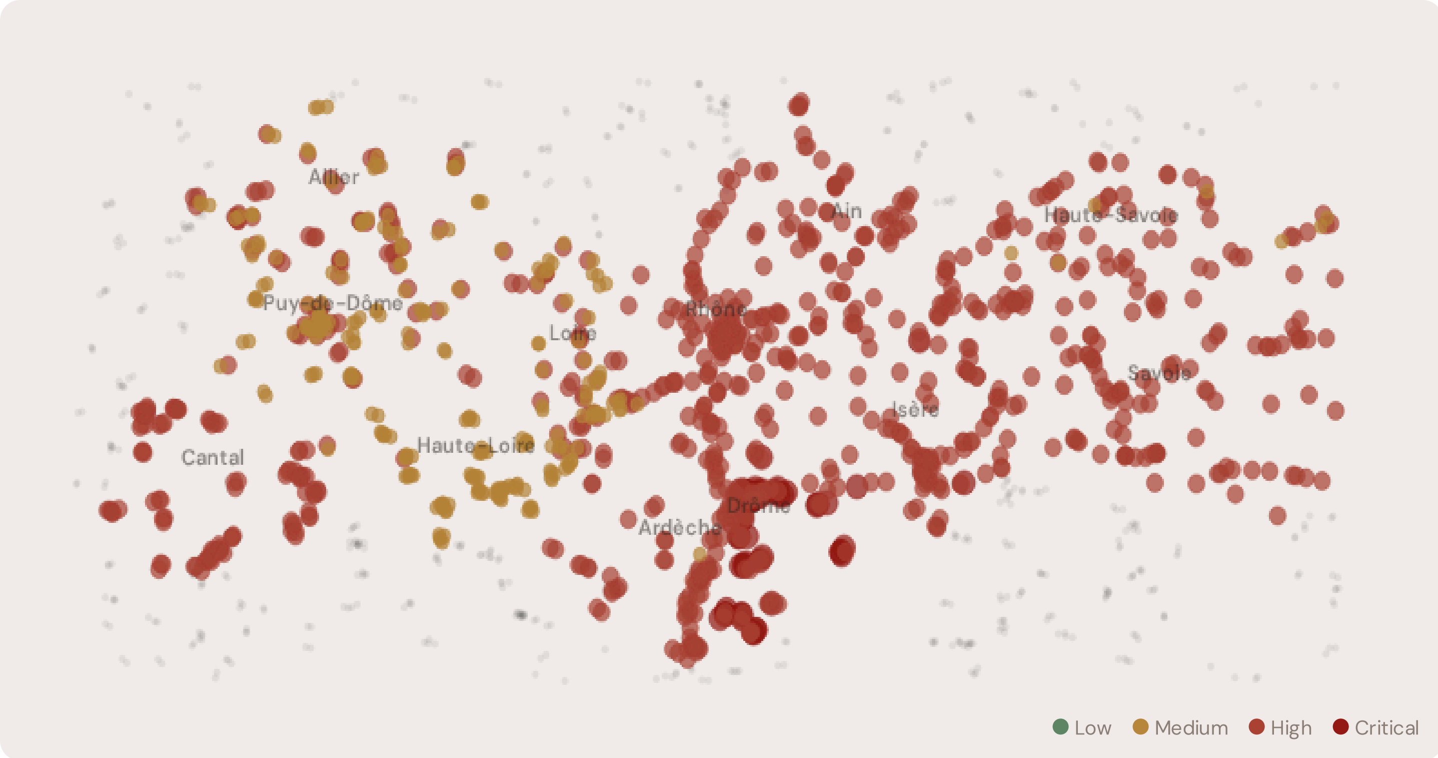
WORST DÉPARTEMENT

Drôme

Avg R: 0.756 · 95 substations

The Rhône Valley hosts **1152 substations** across six départements, making it one of the most energy-critical corridors in Europe. However, its risk profile is disproportionately concentrated in the upper bands: **825 substations (71.6%)** are classified High and **59** Critical, with only **0** in the Low band. 76% of Rhône Valley substations score above the national fleet median of 0.502. The corridor median R of 0.561 reflects the tension between nuclear baseload inflexibility, rapidly growing renewable variability, and aging grid infrastructure built for 1970s load profiles. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under France 2030 timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Poste Drôme 083	Drôme	0.816	Critical	0.629	0.617	0.610	0.268	0.567	0.616	
Poste Drôme 047	Drôme	0.808	Critical	0.663	0.570	0.542	0.265	0.561	0.651	
Poste Drôme 078	Drôme	0.807	Critical	0.643	0.558	0.565	0.272	0.522	0.622	
Poste Drôme 021	Drôme	0.805	Critical	0.614	0.597	0.588	0.235	0.537	0.603	
Poste Drôme 007	Drôme	0.799	Critical	0.656	0.603	0.576	0.332	0.466	0.645	
Poste Drôme 095	Drôme	0.796	Critical	0.648	0.602	0.503	0.321	0.534	0.592	
Poste Drôme 088	Drôme	0.795	Critical	0.614	0.627	0.570	0.277	0.508	0.618	
Poste Drôme 054	Drôme	0.790	Critical	0.621	0.630	0.517	0.282	0.559	0.582	
Poste Drôme 077	Drôme	0.790	Critical	0.643	0.582	0.556	0.245	0.525	0.579	
Poste Drôme 020	Drôme	0.788	Critical	0.560	0.606	0.541	0.226	0.528	0.617	

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
… 1142 additional substations — *explore full corridor in Digital Twin →*										

D.3 Component Analysis

COMPONENT AVERAGES — RHÔNE VALLEY VS FLEET

C — Continuity 0.589 vs 0.497 (+18%)

I — Infrastructure 0.446 vs 0.460 (-3%)

S — Saturation 0.347 vs 0.276 (+26%)

V — Voltage 0.578 vs 0.671 (-14%)

E — Economic 0.338 vs 0.341 (-1%)

T — Transition 0.590 vs 0.568 (+4%)

MODIFIER AVERAGES — RHÔNE VALLEY VS FLEET

R3 Consequence	1.093	fleet 1.091	↑
R4 Graph Crit.	1.012	fleet 1.006	↑
R6a Restoration	0.999	fleet 0.992	→
R6b Seismic	1.034	fleet 1.023	↑
R7 Digital Read.	1.050	fleet 1.014	↑

The dominant risk driver across the Rhône Valley corridor is **T (Transition)**, averaging 0.590 against a fleet average of 0.568 — occupying 1181% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.578 vs fleet 0.671 (578% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.093 (fleet: 1.091), reflecting the socio-economic fragility of The Rhône Valley's industrial-oriented, manufacturing-dependent economy with significant hydroelectric and nuclear bases. The R6b seismic modifier averages 1.034, capturing the region's moderate but non-negligible seismic exposure — a dimension invisible to every competing framework.

D.4 Département Breakdown

PROVINCE RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Drôme 95 subs · avg R = 0.756 · H:39 C:56 M:0

Isère 95 subs · avg R = 0.708 · H:92 C:3 M:0

Savoie 95 subs · avg R = 0.632 · H:95 M:0

Ain 95 subs · avg R = 0.600 · H:95 M:0

Rhône 107 subs · avg R = 0.587 · H:107 M:0

Ardèche 95 subs · avg R = 0.563 · H:94 M:1

Cantal 95 subs · avg R = 0.562 · H:95 M:0

Haute-Savoie 95 subs · avg R = 0.527 · H:88 M:7

Loire 95 subs · avg R = 0.501 · H:45 M:50

Allier 95 subs · avg R = 0.501 · H:48 M:47

Puy-de-Dôme 95 subs · avg R = 0.487 · H:26 M:69

Haute-Loire 95 subs · avg R = 0.464 · H:1 M:94

The province-level breakdown reveals significant intra-regional variation. **Drôme** leads with a mean R of 0.756 across 95 substations, while **Haute-Loire** scores 0.464 — a spread of 0.293 within a single region. This disparity underscores why regional-level metrics (as used by CEER and JRC) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that the Rhône Valley is not uniformly stressed but contains distinct corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

216 Rhône Valley substations score R ≥ 0.650, placing them in the upper quartile of national risk. For PPE (Programmation Pluriannuelle de l'Énergie) investment prioritisation, this analysis identifies the Drôme corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by service-economy dependence, tourism exposure, and above-average energy poverty — are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

European Context Card

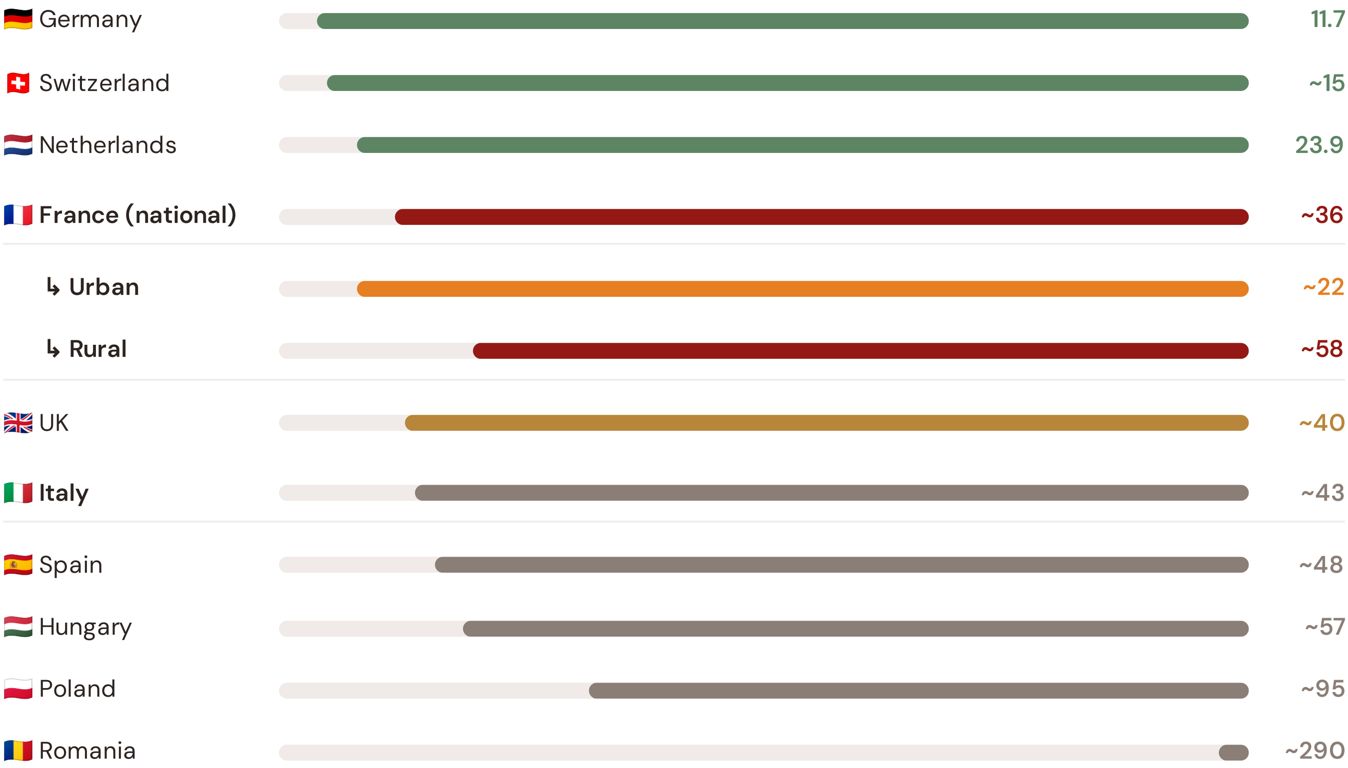
How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Rhône Valley corridor analysis hinges on France's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

EUROPEAN CONTEXT CARD — SAIDI BENCHMARKING

Ed. 001 · April 2026

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



Source: CEER 7th Benchmarking Report (2022); CRE TIQE (2023); Bundesnetzagentur (2024). France sub-categories from CRE territory classification (urban/rural). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

This month's key insight: The Rhône Valley corridor deep-dive shows **1152 substations** (of 1152) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with France's rural territory classification. The Rhône Valley's average C of 0.589 is **1.2× the fleet average** of 0.497. In European terms, these substations individually perform closer to Hungarian or Spanish SAIDI levels (~48–57 min/customer/year) than to France's own national average (~36 min). The SSI reveals what aggregate CEER statistics conceal: that France's mid-tier European position is not a national condition but a statistical average of Dutch-quality urban performance and Eastern-European-level rural performance — and the Rhône Valley corridor concentrates the worst of the latter.

Regional Risk Ranking — All 13 French Régions

Where does the Rhône Valley sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

#1 Provence-Alpes-Côte d'Azur — 562 subs · avg R = 0.638 · H:559 C:3

#2 Corse — 32 subs · avg R = 0.628 · H:32

#3 Occitanie — 973 subs · avg R = 0.609 · H:879 C:17

#4 **Auvergne-Rhône-Alpes** 1152 subs · avg R = 0.574 · H:825 C:59

#5 Nouvelle-Aquitaine 902 subs · avg R = 0.568 · H:839

#6 Bretagne 280 subs · avg R = 0.523 · H:224

#7 Pays de la Loire 356 subs · avg R = 0.514 · H:187

#8 Hauts-de-France 522 subs · avg R = 0.497 · H:189

#9 Centre-Val de Loire 360 subs · avg R = 0.477 · H:51

#10 Normandie 358 subs · avg R = 0.472 · H:105

#11 Grand Est 765 subs · avg R = 0.445 · H:51

#12 Bourgogne-Franche-Comté 520 subs · avg R = 0.398 · H:0

#13 Île-de-France 1116 subs · avg R = 0.258 · H:0

The Rhône Valley ranks **#4 of 13 régions** by mean R_median, placing it among the upper tier of national risk. The three most stressed regions are Provence-Alpes-Côte d'Azur, Corse, Occitanie, while the three most resilient are Île-de-France, Bourgogne-Franche-Comté, Grand Est. 8 of 13 régions score above the fleet average of 0.495. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress across France. It is precisely this substation-level granularity that positions the SSI as a tool for PPE / France 2030 / France Relance investment prioritisation: not treating entire régions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the Enedis/RTE and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	Enedis/RTE + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.6 · Inaugural edition · April 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. A BESS developer considering a €10M commitment needs to understand not just that a substation scores highly, but which specific factors drive that score — and whether those factors are likely to persist. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history — and can verify this against CRE (Commission de Régulation de l'Énergie)'s published TIQE data. Transparency is not optional; it is what converts model outputs into investment decisions.

KEY CONCEPTS

→ SHAP: game-theoretic feature attribution per prediction

→ Waterfall visualisation: fleet average → substation prediction

→ Direct traceability to SSI components and data sources

→ Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The French training set comprises 7,898 substations with complete 95-feature vectors. Average CI_width of 0.188 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 397 Low, 3481 Medium, 3941 High, 79 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 002

The PACA Coastal Grid

Deep-dive into Provence-Alpes-Côte d'Azur's grid risk profile — substation map, component analysis, province breakdown. European Context Card: Mediterranean thermal stress trajectory vs Atlantic peers.

NEXT DATA REFRESH

Q3 2026 — 2 Quarterly Sources

SDES, Bpifrance are due for refresh in Q3. Impact: ODRÉ (Registre National des Installations) data refresh schedule (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **CRE (Commission de Régulation de l'Énergie)** (8 variables → C1–C4 (SAIDI/SAIFI), I1–I3 (IRI), quality regulation).

FLEET SPOTLIGHT

0 Substations Approaching Critical Degradation

The SSI's 4-state Markov degradation model identifies **0 substations** with a stationary critical probability above 70% — meaning their long-term equilibrium state is dominated by degraded or critical condition. Without intervention, these substations will trend toward failure. Average estimated time-to-critical: ? years.

Edition 002 will be published on the second Thursday of May 2026. Deep-dive region: **Provence-Alpes-Côte d'Azur**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a French utility, CRE, RTE, Enedis, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 001 (Inaugural)

Published 9 April 2026 · SSI Index v4.0.2 · 7,898 substations · ~12,400 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5
